WEEKLY ASSIGNMENT - 1

Database Creation:

```
Create database salesDB;
use salesDB;
```

Tables Creation:

```
-- CREATE CLIENT TABLE

CREATE TABLE Client (
    ClientNo VARCHAR(6) PRIMARY KEY,
    Name VARCHAR(20) NOT NULL,
    Address1 VARCHAR(30),
    Address2 VARCHAR(30),
    City VARCHAR(15),
    PinCode NUMERIC(8),
    State VARCHAR(15),
    BalDue DECIMAL(10,2),
    CHECK (ClientNo LIKE 'C%')
);

-- CREATE PRODUCT TABLE

CREATE TABLE Product (
    ProductNo VARCHAR(6) PRIMARY KEY,
    Description VARCHAR(15) NOT NULL,
    ProfitPerc DECIMAL(4,2) NOT NULL,
    UnitMeasure VARCHAR(10) NOT NULL,
    QtyOnHand NUMERIC(8) NOT NULL,
    ReorderLvl NUMERIC(8) NOT NULL,
    SellPrice DECIMAL(8,2) NOT NULL,
    CostPrice DECIMAL(8,2) NOT NULL,
    CHECK (ProductNo LIKE 'P%'),
    CHECK (SellPrice <> 0),
    CHECK (CostPrice <> 0)
);

-- CREATE SALESMAN TABLE

CREATE TABLE Salesman (
    SalesmanNo VARCHAR(6) PRIMARY KEY,
    SalesmanName VARCHAR(20) NOT NULL,
    Address1 VARCHAR(30) NOT NULL,
    Address2 VARCHAR(30),
    City VARCHAR(20),
    PinCode NUMERIC(8),
    State VARCHAR(20),
    SalAmt DECIMAL(8,2) NOT NULL,
    TgtToGet DECIMAL(6,2) NOT NULL,
    YtdSales DECIMAL(6,2) NOT NULL,
    Remarks VARCHAR(60),
    CHECK (SalesmanNo LIKE 'S%'),
    CHECK (SalAmt <> 0)
);

-- CREATE SALES ORDER TABLE
```

```sql
CREATE TABLE SalesOrder (
    OrderNo VARCHAR(6) PRIMARY KEY,
    ClientNo VARCHAR(6),
    OrderDate DATE,
    DelyAddr VARCHAR(25),
    SalesmanNo VARCHAR(6),
    DelyType CHAR(1),
    BilledYN CHAR(1),
    DelyDate DATE,
    OrderStatus VARCHAR(20),
    CHECK (OrderNo LIKE 'O%'),
    CHECK (DelyType IN ('P','F')),
    CHECK (BilledYN IN ('Y','N')),
    CHECK (OrderStatus IN ('In Process','Fulfilled','Backorder','Cancelled')),
    FOREIGN KEY (ClientNo) REFERENCES Client(ClientNo),
    FOREIGN KEY (SalesmanNo) REFERENCES Salesman(SalesmanNo)
);

-- CREATE SALES ORDER DETAILS TABLE

CREATE TABLE SalesOrderDetails (
    OrderNo VARCHAR(6),
    ProductNo VARCHAR(6),
    QtyOrdered NUMERIC(8),
    QtyDisp NUMERIC(8),
    ProductRate DECIMAL(10,2),
    PRIMARY KEY (OrderNo, ProductNo),
    FOREIGN KEY (OrderNo) REFERENCES SalesOrder(OrderNo),
    FOREIGN KEY (ProductNo) REFERENCES Product(ProductNo)
);
```

Inserting Records:

```sql
-- INSERT RECORDS TO CLIENT TABLE
INSERT INTO Client VALUES
('C0001','Amit','MG Road','Andheri','Mumbai',400001,'Maharashtra',1200.50),
('C0002','Neha','Link Road','Bandra','Mumbai',400050,'Maharashtra',500.00),
('C0003','Rahul','Park Street','Salt Lake','Kolkata',700001,'West Bengal',0.00),
('C0004','Sneha','Ring Road','Indiranagar','Bangalore',560001,'Karnataka',800.75),
('C0005','Kiran','Main Street','Abids','Hyderabad',500001,'Telangana',300.00),
('C0006','Ivan Bayross','Hill
Road','Bandra','Mumbai',400050,'Maharashtra',2500.00);

-- INSERT RECORDS TO PRODUCT TABLE
INSERT INTO Product VALUES
('P0001','1.44 drive',10.50,'Piece',100,20,1150.50,900),
('P0002','Trousers',15.00,'Piece',150,30,2500,2000),
('P0003','Pull Overs',12.00,'Piece',200,40,1800,1200),
('P0004','Monitor',20.00,'Piece',80,15,5200,4500),
('P0005','Printer',18.00,'Piece',60,10,6000,5000);

-- INSERT RECORDS TO SALESMAN TABLE
INSERT INTO Salesman VALUES
('S0001','Aman','Park
Ave','Ameerpet','Hyderabad',500016,'Telangana',25000,100,50,'Good'),
('S0002','Anil','Main Road','Banjara
Hills','Hyderabad',500034,'Telangana',22000,120,4500,'Excellent'),
```

```sql
('S0003','Meena','MG
Road','Brigade','Bangalore',560025,'Karnataka',18000,90,3000,'Average');

-- INSERT RECORDS TO SALES ORDER TABLE
INSERT INTO SalesOrder VALUES
('O19001','C0001','2002-04-10','Mumbai','S0001','P','Y','2026-01-04','Fulfilled'),
('O19002','C0002','2024-01-12','Mumbai','S0002','F','N','2024-01-18','In
Process'),
('O19003','C0003','2024-01-20','Bangalore','S0003','P','Y','2024-01-
25','Fulfilled'),
('O19004','C0006','2026-01-05','Mumbai','S0001','P','Y','2026-01-15','Fulfilled');

-- INSERT RECORDS TO SALES ORDER DETAILS TABLE
INSERT INTO SalesOrderDetails VALUES
('O19001','P0001',10,8,1200.00),
('O19001','P0002',5,5,2500.00),
('O19002','P0001',2,2,1200.00),
('O19002','P0003',3,10,1800.00),
('O19003','P0004',6,6,5200.00),
('O19003','P0005',3,2,6000.00),
('O19004','P0001',2,2,1200.00),
('O19004','P0002',1,1,2500.00);
```

Answer following queries with the help of above schema:

```sql
-- 1. Display the names of all the clients.

      select Name from Client;
```

	Name
1	Amit
2	Neha
3	Rahul
4	Sneha
5	Kiran
6	Ivan Bayross

```sql
-- 2. Display all the clients who are located in Mumbai.

      select * from Client where City='Mumbai';
```

	ClientNo	Name	Address1	Address2	City	PinCode	State	BalDue
1	C0001	Amit	MG Road	Andheri	Mumbai	400001	Maharashtra	1200.50
2	C0002	Neha	Link Road	Bandra	Mumbai	400050	Maharashtra	500.00
3	C0006	Ivan Bayross	Hill Road	Bandra	Mumbai	400050	Maharashtra	2500.00

```sql
-- 3. Display all the products whose selling price is > 2000 and < 5000.

      select * from Product where SellPrice between 2000 and 5000;
```

	ProductNo	Description	ProfitPerc	UnitMeasure	QtyOnHand	ReorderLvl	SellPrice	CostPrice
1	P0002	Keyboard	15.00	Piece	150	30	2500.00	2000.00

```sql
-- 4. Display Name, City and State of Clients not in the state of Maharashtra.

select Name,City,State from Client where state not in ('Maharashtra');
```

	Name	City	State
1	Rahul	Kolkata	West Bengal
2	Sneha	Bangalore	Karnataka
3	Kiran	Hyderabad	Telangana

```sql
-- 5. Display all the information of client no C0001 and C0002.

select * from Client where ClientNo='C0001' or ClientNo='C0002';
```

	ClientNo	Name	Address1	Address2	City	PinCode	State	BalDue
1	C0001	Amit	MG Road	Andheri	Mumbai	400001	Maharashtra	1200.50
2	C0002	Neha	Link Road	Bandra	Mumbai	400050	Maharashtra	500.00

```sql
-- 6. Change the selling price of '1.44 drive' to Rs. 1150.50.

update Product
set SellPrice=1150.50 where Description='1.44 drive';

-- 7. Delete the record of client no C0005.

delete from Client where ClientNo ='C0005';

-- 8. Display the clients who stay in a city whose second letter is 'a'.

select * from Client where City like'_a%';
```

	ClientNo	Name	Address1	Address2	City	PinCode	State	BalDue
1	C0004	Sneha	Ring Road	Indiranagar	Bangalore	560001	Karnataka	800.75

```sql
-- 9. Count the number of products having price greater than or equal to 1500.

select count(*) as no_of_products from Product where SellPrice>=1500;
```

	no_of_products
1	4

```sql
-- 10. Display qtyordered, qtydisp and balanceqty (not in table).

select s.QtyOrdered,s.QtyDisp,p.QtyOnHand as balanceQty from
SalesOrderDetails as s join product as p
on s.ProductNo=p.ProductNo;
```

	QtyOrdered	QtyDisp	balanceQty
1	10	8	100
2	5	5	150
3	2	2	100
4	3	10	200
5	6	6	80
6	3	2	60
7	2	2	100
8	1	1	150

Write Commands to do Following:

```sql
-- 1. Make Client_no as primary key in client_master.

      ALTER TABLE Client
      ADD CONSTRAINT PK_Client PRIMARY KEY (ClientNo);

-- 2. Add a new column phone_no in the client_master table.

      ALTER TABLE Client
      ADD PhoneNo VARCHAR(15);

-- 3. Add the not null constraint in the product_master table with the column
description, profit percent, sell price and cost price.

      ALTER TABLE Product
      ALTER COLUMN Description VARCHAR(15) NOT NULL;

      ALTER TABLE Product
      ALTER COLUMN ProfitPerc DECIMAL(5,2) NOT NULL;

      ALTER TABLE Product
      ALTER COLUMN SellPrice DECIMAL(8,2) NOT NULL;

      ALTER TABLE Product
      ALTER COLUMN CostPrice DECIMAL(8,2) NOT NULL;


-- 4. Change size of name column to 60 in client_master table.

      ALTER TABLE Client
      ALTER COLUMN Name varchar(60);

-- 5. Remove pincode column from table.

      ALTER TABLE Client
      DROP COLUMN PinCode;
```

Define in 1 or 2 lines and give one example also:

1. Recursive Relationship

A recursive relationship is when a table is related to itself.
Example: In a Salesman table, if a salesman is supervised by another salesman in the same table, it forms a recursive relationship.

2. Composite Key

A composite key is a primary key made using two or more columns together.
Example: (OrderNo, ProductNo) in the SalesOrderDetails table.

3. The LIKE Operator with Pattern Matching

The LIKE operator is used to search for a specific pattern in a column.

Example: select * from Client where City like 'M%';

4. DROP TABLE Command

The DROP TABLE command is used to permanently remove a table from the database.
Example: drop table Product;

5. FULL OUTER JOIN

A FULL OUTER JOIN returns all rows from both tables whether a match exists or not.
Example:
select Client.ClientNo, SalesOrder.OrderNo
from Client
full outer join SalesOrder
on Client.ClientNo = SalesOrder.ClientNo;

```
-- WRITE QUERIES FOR FOLLOWING DESCRIPTIONS: (Joins)

-- 1. Find out the products which have been sold to 'Ivan Bayross'.

      select distinct p.productno, p.description from client c
      join salesorder s
      on c.clientno = s.clientno
      join salesorderdetails sd
      on s.orderno = sd.orderno
      join product p
      on sd.productno = p.productno
      where c.name = 'Ivan Bayross';
```

	productno	description
1	P0001	1.44 drive
2	P0002	Trousers

```
-- 2. Finding out the products and their quantities that will have to be delivered
in the current month.

      select
      p.productno,
      p.description,
      sd.qtyordered
      from salesorder s
      join salesorderdetails sd
      on s.orderno = sd.orderno
      join product p
      on sd.productno = p.productno
      where month(s.delydate) = month(getdate()) and year(s.delydate) =
year(getdate());
```

	productno	description	qtyordered
1	P0001	1.44 drive	10
2	P0002	Trousers	5
3	P0001	1.44 drive	2
4	P0002	Trousers	1

```
-- 3. Listing the ProductNo and description of constantly sold (i.e. rapidly
moving) products.

	select p.productno, p.description
	from product p
	join salesorderdetails sd on p.productno = sd.productno
	group by p.productno, p.description
	having count(distinct sd.orderno) > 1;
```

	productno	description
1	P0001	1.44 drive
2	P0002	Trousers

```
-- 4. Finding the names of clients who have purchased 'Trousers'.

	select distinct c.clientno, c.name from client c
	join salesorder s
	on c.clientno = s.clientno
	join salesorderdetails sd
	on s.orderno = sd.orderno
	join product p
	on sd.productno = p.productno
	where p.description = 'Trousers';
```

	clientno	name
1	C0001	Amit
2	C0006	Ivan Bayross

```
-- 5. Listing the products and orders from customers who have ordered less than 5
units of 'Pull Overs'.

	select
	s.orderno,
	c.name as client_name,
	p.productno,
	p.description,
	sd.qtyordered
	from client c
	join salesorder s
	on c.clientno = s.clientno
	join salesorderdetails sd
	on s.orderno = sd.orderno
	join product p
	on sd.productno = p.productno
	where p.description = 'Pull Overs'and sd.qtyordered < 5;
```

	orderno	client_name	productno	description	qtyordered
1	O19002	Neha	P0003	Pull Overs	3

```
-- WRITE QUERIES FOR FOLLOWING DESCRIPTIONS: (Subqueries)

-- 1. Finding the non-moving products i.e. products not being sold.

	select * from product
	where productno not in (
```

```
select distinct productno from salesorderdetails);
```

	ProductNo	Description	ProfitPerc	UnitMeasure	QtyOnHand	ReorderLvl	SellPrice	CostPrice

```
-- 2. Finding the name and complete address for the customer who has placed Order
number 'O19001'.

select name, address1, address2, city, pincode, state
from client
where clientno =(
select clientno from salesorder where orderno = 'O19001');
```

	name	address1	address2	city	pincode	state
1	Amit	MG Road	Andheri	Mumbai	400001	Maharashtra

```
-- 3. Finding the clients who have placed orders before the month of May '02.

select * from client where clientno in (
select clientno from salesorder
where orderdate < '2002-05-01');
```

	ClientNo	Name	Address1	Address2	City	PinCode	State	BalDue
1	C0001	Amit	MG Road	Andheri	Mumbai	400001	Maharashtra	1200.50

```
-- WRITE COMMANDS TO DO FOLLOWING

-- 1. Display system date as Saturday, February 11, 2012.

select format(cast('2012-02-11' as date), 'dddd, MMMM dd, yyyy') as
SystemDate;
```

	SystemDate
1	Saturday, February 11, 2012

```
-- 2. Display Balance Due from Client master as $99,999.99.

select name, format(baldue, '$99,999.99') as BalanceDue from client;
```

	name	BalanceDue
1	Amit	$99999120199
2	Neha	$9999950099
3	Rahul	$9999999
4	Sneha	$9999980199
5	Ivan Bayross	$99999250099

```
-- 3. Display message as 'Salesman Aman sold goods of 50 while given target was
100'.

select
'Salesman ' + SalesmanName +
'sold goods of ' + cast(YtdSales as varchar) +
' while given target was ' + cast(TgtToGet as varchar) as Message
from Salesman where SalesmanName = 'Aman';
```

	Message
1	Salesman Amansold goods of 50.00 while given target was 100.00

```sql
-- 4. Display your Age in Years.

select datediff(year, '2005-03-20', getdate()) as AgeInYears;
```

	AgeInYears
1	21